Nancy **Sotero Silva**

NEUROSCIENCE RESEARCHER · UNIVERSITÄT BIELEFELD

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"The mind that opens to a new idea never returns to its original size"

Summary

Audiology and Neuroscience researcher who seeks to use technology to understand how the brain processes the information received by the ears, why sometimes this fails and how we can overcome these impairments. My background includes training in rehabilitation and research experience, with a keen interest in cognition, communication, perception and performance. I am currently working on the integration between the oculomotor and auditory efferent systems, but have also trained in cortical and behavioral measurements.

Education

Bielefeld University — *Bielefeld, Germany*

PHD IN BIOLOGY - AREA: COGNITIVE NEUROSCIENCE — *July 2023 - July 2026*

- Dissertation: Integration of oculomotor signals into the auditory periphery.
  Supervisors: Prof. Dr. Christoph Kayser and Dr. Norbert Böddeker

Santos Dumont Institute — *Macaíba, Brazil*

MSC IN NEUROENGINEERING; RESIDENCY ON HEALTH CARE OF THE PERSON WITH DISABILITY — *March 2019 - February 2023*

- Master's thesis: Influence of stimulus complexity on cortical electrophysiological responses in adults with normal hearing through multichannel acquisition.
  Supervisor: Prof. Dr. Edgard Morya

Federal University of Paraíba — *João Pessoa, Brazil*

BSC IN SPEECH, LANGUAGE AND HEARING SCIENCES — *April 2014 - July 2018*

- Bachelor's thesis: "Absolute Pitch" in music graduation students: is there a relationship with temporal auditory processing?
  Supervisors: Prof. Dr. Hannalice Gottschalck Cavalcanti and Dr. Raphael Bender Chagas Leite

Research Experince

Cognitive Neuroscience Lab, Bielefeld University — *Bielefeld, Germany*

SCIENTIFIC WORKER — *July 2023 - Currently*

- Conducts in-ear recordings (using BioSemi EEG system), eye tracking (EyeLink), hearing tests and psychophysics experiments (PsychToolbox); as long as data curation, processing (FieldTrip), analysis and reporting.

Medical Image Processing Lab, EPFL — *Geneva, Switzerland*

MASTER'S RESEARCH INTERNSHIP — *July - August 2022*

- 2 months internship - Summer Research Program.
- Explored processing pipeline optimization (MATLAB) for fNIRS data (Artinis) elicited by cognitive and auditory stimuli. Inspectioned data from open databases, tested fNIRs processing tools, prepared presentation and reports.

Santos Dumont Institute — *Macaíba, Brazil*

MASTER'S AND RESIDENCY RESEARCH PROJECT — *2019 - 2023*

- Project: Electrophysiological, vascular/hemodynamic and pupillometric measures in cortical plasticity monitoring of adult hearing aids users.
- Performed experiment design, data collection and analysis with electroencephalography (BrainVision), ABR, FFR, pupillometry (Mangold) and fNIRS (NirX), clinical assessment and follow-up of patients.

Federal University of Paraíba — *João Pessoa, Brazil*

BACHELOR'S RESEARCH PROJECT — *2017 - 2018*

- Identified the prevalence of absolute pitch among music undergraduates and its relation to auditory temporal resolution assessment tasks.
- Performed psychophysical experimentation, central auditory processing assessment and music cognition tests; as well as written reports and presentations.

OTHER RESEARCH AND COMMUNITY OUTREACH EXPERIENCES — *2015 - 2021*

- Communication and hearing assessment and rehabilitation for low-income patients, planning and execution of educational events for the community, organisation of scientific events, reports and presentations.
  - Prevalence of absolute pitch among music undergraduates and its relation to auditory temporal resolution
  - Transdisciplinary assessment of patients complaining of tinnitus
  - Speech therapy for patients with fluency disorders
  - Impact of noise on workers' health

Teaching and Supervision

2021-2022	**Temporary Lecturer - UFRN**, Otoneurology, Balance Disorders and Occupational Audiology courses and supervision in Hearing Rehabilitation internships, bachelor students.	*Natal, Brazil*
2024-2026	**Tutor/Lectures - Bielefeld University**, Multisensory Perception and Perception and Action modules, bachelor and master's students.	*Bielefeld, Germany*

Skills

- **Research** Data acquisition and analysis: Electrophysiology (EEG, ERPs), fNIRS, Eyetracking (Pupillometry, Saccades). Programming: Python (basic), MATLAB (intermediate), LaTeX. Scientific Writing.
- **Languages** Proficient in English, Portuguese (native), and Spanish. Basic knowledge of German and French.

Music practice and instruction

2018-2020	**BSc - interrupted**, Degree in music - habilitation in classical singing	*João Pessoa, Brazil*
2018-2019	**Research**, Higher education in music: analysis of music undergraduation programs in Brazilian Northeast	*João Pessoa, Brazil*
2016-2018	**Performance**, Gazzi de Sá University Choir [choir and soloist]	*João Pessoa, Brazil*
2017-2018	**Performance**, Presto Ópera - classical singing for the community	*João Pessoa, Brazil*
2014-2015	**Performance and community outreach**, Speech Therapy Students' Choir	*João Pessoa, Brazil*

Awards, Honors and Stipends

2026	**Cambridge Open Lab**, Stipend for attendance	*Cambridge, UK*
2025	**CNSP Hackathon**, Awarded 2nd best participant - fNIRS Hyperscanning	*Maastricht, The Netherlands*
2025	**10th ISAAR, William Demant Foundation**, Stipend for attendance and presentation	*Nyborg, Denmark*
2024	**FENS, IBRO-PERC - Early Career Program Award**, Stipend for a 3-weeks internship	*Budapest, Hungary*
2024	**EPFL - Summer Research Program**, Stipend for a 2-months internship	*Geneva/Lausanne, Switzerland*
2019-2021	**Ministry of Health, Federal Government of Brazil**, Postgraduate Residency Stipend (24 months)	*Natal, Brazil*
2018	**Federal University of Paraíba**, Awarded "Academic Laurea" (graduation with honours) in BSc	*João Pessoa, Brazil*
2016-2018	**Ministry of Education, Federal Government of Brazil**, Stipends for teaching assistantship (12 months) and extracurricular internship (12 months) during BSc	*João Pessoa, Brazil*

Presentations - Selected

Oral presentations

2026	**24th IMRF - International Multisensory Research Forum**, Exploring the role of auditory efferences in EMREOs [Young Researcher Talk Session]	*Genoa, Italy*
2025	**47th ECVP - European Conference on Visual Perception**, The EMREOs time course differs between ipsi- and contralateral saccades	*Mainz, Germany*
2023	**9th Neuroengineering Symposium**, Comparing different stimulation complexities on LLAEPs for temporal and central cortical areas in adults	*Macaíba, Brazil*
2022	**3rd VCCA - Virtual Conference on Computational Audiology**, Electrophysiological and pupillometric measures in monitoring auditory cortex plasticity and listening effort post hearing aids	*Virtual*
2022	**Rehabilitation Colloquium, McGill University**, SARS-CoV-2 infection and audiological findings	*Virtual*

Posters

2026	**FENS Forum**, Consistency of EMREOs signals across multiple experiments: evidence for a highly conserved oculomotor connection [presenter]	*Barcelona, Spain*
2026	**Vision Augmented Hearing, Royal Society**, EMREOs during noise and head-tilts [presenter]	*Manchester, UK*
2025	**8th ICAC - International Conference on Auditory Cortex**, Relation of EMREOs amplitude and time course with saccade direction and tympanometric measurements [presenter]	*Maastricht, The Netherlands*
2025	**10th ISAAR - International Symposium on Auditory and Audiological Research**, EMREOs repercussions on auditory spatial localization [presenter]	*Nyborg, Denmark*
2024	**FENS Forum**, EMREOs are induced by both visual- and auditory-guided saccades [presenter]	*Vienna, Austria*
2023	**38th EIA - International Audiology Meeting**, Pupillometric measurements and their relationship to auditory effort & Cognitive evoked potential in normal-hearing adults: active and passive oddball paradigms	*Florianópolis, Brazil*
2022	**8th Neuroengineering Symposium**, fNIRS data elicited by motor or auditory stimuli: Processing pipeline and tools for analysis feature extraction [presenter]	*Macaíba, Brazil*

- Between 2013-2023: around five other oral presentations and 30 posters.

Publications - Selected

1. Sotero Silva, N., & Kayser, C. (2026). **Stability of Eye Movement-Related Eardrum Oscillations to acoustic and gravitational manipulations**. bioRxiv, (preprint), `https://doi.org/10.64898/2026.04.16.718961` (currently under review at Plos One)

2. Sotero Silva, N., Bröhl, F. & Kayser, C. (2026) **Sound lateralization ability is affected by saccade direction but not EMREOs**. Journal of Neurophysiology, 135(5), `https://doi.org/10.1152/jn.00555.2025`

3. Wilroth, J., Sotero Silva, N., Tafakkor, A., Mesquita, B., Lau, B., Ip, E., Hannah, J. & Di Liberto, G. (2026). **fNIRS for continuous speech: considerations for hyperscanning**. bioRxiv (preprint), `https://doi.org/10.64898/2026.03.20.713212` (currently under review at Hearing Research)

4. Sotero Silva, N., Kayser, C. & Bröhl, F. (2025). **Unraveling eye movement-related eardrum oscillations (EMREOs): how saccade direction and tympanometric measurements relate to their amplitude and time course**. Hearing Research, 467. `https://doi.org/10.1016/j.heares.2025.109276`

5. Cavalcanti, H.G., Lucena, M.H.M.S.L., Morya, E., Sotero Silva, N. & Paiva S.F. (2024) **Interface between Music and Audiology** (Chapter). In: Delgado, I.C., Andrade, K.C.L. & Rosa, M.R.D. (ed.). Speech Therapy Intervention in Hearing and Language. 1ed. Rio de Janeiro: Thieme Revinter, p. 56-66. ISBN: 978-65-5572-283-3 [available in portuguese]

6. Sotero Silva, N., Evangelista, C.K.S., Cavalcanti, P.F., Balen, S.A., Cavalcanti, H.G. & Morya, E. (2022). **Review of Audiological Findings in SARS-CoV-2 Infection**. Journal of Neurological Disorders, 10(8). `https://doi.org/10.4172/2329-6895.10.9.514`

- Between 2020-2022: three other papers and one book chapter on Occupational Audiology and Tinnitus [available in portuguese].

Manuscripts in preparation

1. Sotero Silva, N., Vasconcelos, I.C., Alves, E.R.A., Monteiro, T.A., Spinelli, B.G., Miranda, D.C.S., Canto-Pereira, V.R., Balen, S.A. & Morya, E. (2026) **Auditory evoked potentials: does stimulus complexity influence responses?** In preparation

2. Campos, L.G.N., Sotero Silva, N., Alves, E.R.A., Silva, A.R.X., Soares, A.K.A., Dias, R.C.T., Brasil, F.L., Morya, E. & Rosa MRD. (2026) **Transcranial direct current stimulation reduces tinnitus discomfort**. In preparation.

3. Alexandre, V.G.A., Sotero Silva, N. & Vasconcelos, L.D.S. (2026) **Neuropsychology in speech therapy** (Chapter). In: Pinto, P.V.N., Vasconcelos, L.D.S. & Silva, F.T.M. (ed.) History of speech therapy in Brazil. 1. ed. Carapicuíba: Pró-fono. In press. [available in portuguese]